

Leonardo Zavala-Jimenez

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EDUCATION

University of California, Davis

Expected June 2026

Bachelor of Science in Computer Science, Statistics Minor

- Coursework: **Operating Systems**, Algorithm Design & Analysis, Intro to **Applied Cybersecurity**, Computer Architecture, Applied Statistics, **Software Engineering**, **Computer Networks**
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SKILLS

Languages: Python, R, Java, Javascript, C, C++, TypeScript, Rust, Swift

Frameworks & Tools: React, Node.js, SQL/SQLite, GitHub, Tailwind CSS, Tableau, Prisma ORM, Resend, MongoDB, ESP-IDF

Specializations: Data Analysis, Machine Learning, Software Engineering, AI

Soft Skills: Team Leadership, Mentorship, Cross-Functional Collaboration, Exemplary Communication

PROJECTS

Software Engineer — Department of Internal Medicine at UC Davis

Jan 2026 — Present

- **Architecting** a full-stack funding workflow platform in **Next.js** and **TypeScript**, projected to eliminate **250+ hours of manual processing annually** by digitizing **500+ departmental submissions**.
- **Engineered relational data models** with **Prisma ORM** and **MongoDB** to manage funding requests, multi-stage approval states, and full revision history with **end-to-end type safety**.
- **Built a tokenized, email-based approval system** via **Resend**, enabling asynchronous multi-reviewer workflows and cutting manual status follow-ups to zero.
- **Enforced** strict runtime data validation using **Zod** schemas, eliminating an entire class of **type-mismatch bugs** across API boundaries.

Software Lead — MuscleMaxx Startup

Jan 2025 — Present

- **Spearheading** development of a cross-platform fitness app in **React Native** and **TypeScript**, integrating a custom **EMG + IMU wearable** built on an **ESP32** microcontroller with an **Adafruit 9-DoF IMU** to deliver real-time muscle activity and motion insights.
- **Designed and implemented a BLE communication layer** on **ESP32** and on the client in **react-native-ble-plx**, enabling **low-latency streaming** of EMG and IMU signals from hardware to the mobile app.
- **Architected a real-time data pipeline** with **SQLite** and **Node.js** to support **live EMG/IMU signal graphing**, sensor-fusion post-processing, and persistent session history.
- **Directing a 5-person cross-functional team** across software, hardware, and business to align delivery milestones with product and go-to-market goals.

GitHub Data Fetcher — Software Engineering Class

Sept 2025 — Nov 2025

- Developed a **Rust CLI tool** that interacts with GitHub **REST API** to retrieve repository and issue metadata, using **SOLID** principles in the architecture.
 - Stored structured JSON objects in **Redis** for fast retrieval and caching, improving query efficiency and reducing API calls by **80%**
 - Applied **asynchronous programming**, **error handling**, and **modular design** to build a reliable and efficient backend data pipeline
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LEADERSHIP

National Outreach Lead / Project Manager, ColorStack @ UC Davis

Dec 2023 — Present

- Represent UC Davis in national ColorStack meetings; maintain documentation and reporting
- Planned events increasing membership by **30%**, supporting underrepresented students in tech
- Overseeing **website creation progress & GitHub repository**, improving members' technical collaboration